SIME DARBY PROPERTY · UNIVERSITY HACKATHON 2026

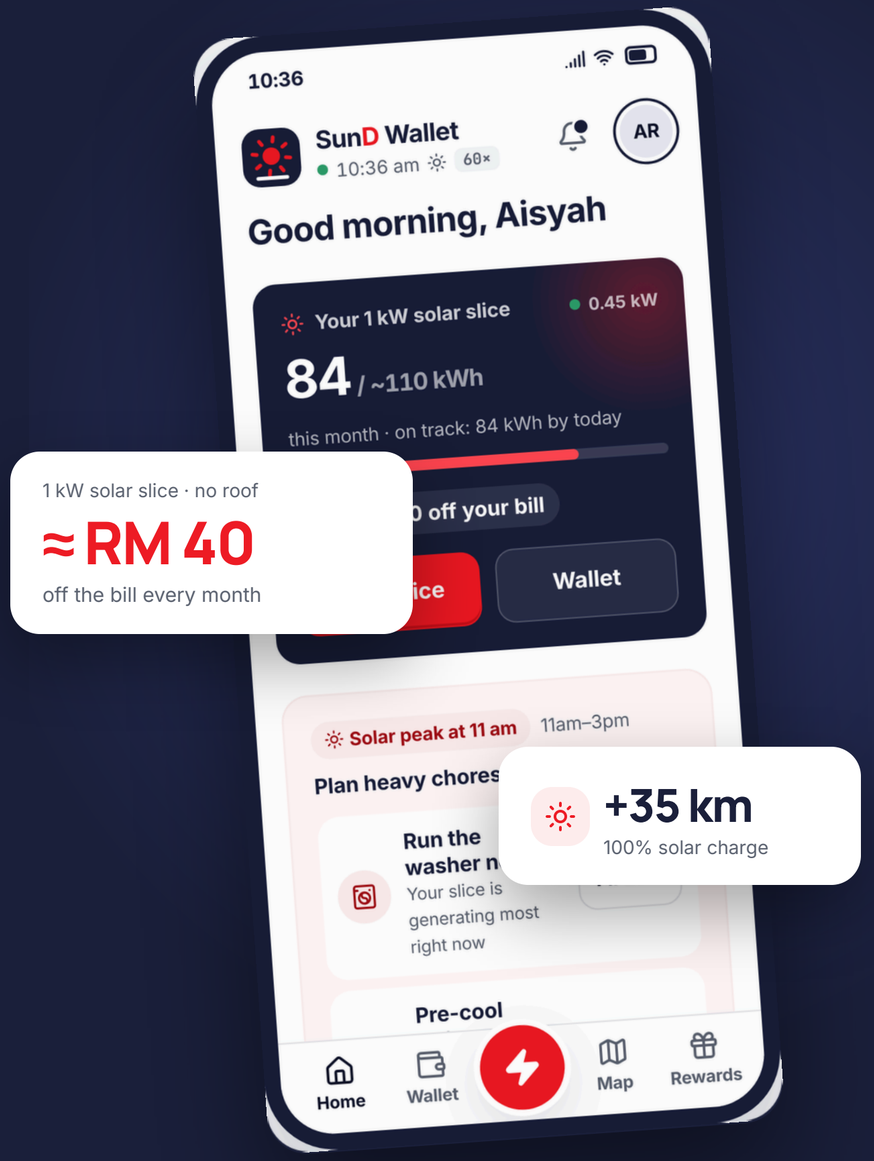
TOWNSHIP OF TOMORROW · PLANET

# SunD

## The sun is already on our roofs.
## Let’s put it in residents’ hands.

**SunD Wallet + SunD Bay**: one wallet turns township solar into credits; solar-first bays turn them into cheaper, cleaner charging.

- Energy efficiency
- Green transport
- Circular economy
- Climate resilience




### TEAM URBANBYTE

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# Sime Darby townships already make solar, but residents can't capture it, spend it or feel it.

CAPTURE · NO ROOF

## 22.8%

**of Malaysian homes are high-rise: no roof for solar**

37% in Selangor. Every rooftop scheme needs a roof.

Landed ✓ ~~Condo~~ ~~Renter~~ ~~B40~~

SPEND · NO CLEAN CHARGING

## 18 : 1

**battery EVs for every public charging bay**

122,286 BEVs vs 6,904 bays; the grid is 92% fossil.



Peninsular power 2025 · coal 58.6% · gas 33.1% · other 8.3%

FEEL · NO ESCAPE FROM HEAT

## ~1M

**homes pushed past 600 kWh by the heat in 2026**

Higher bills, and more time stuck indoors.

Each +1 °C raises elderly hospital admissions in Selangor by **12.9%**

INSIDE SIME DARBY TOWNSHIPS

## 0 kWh

**of Elmina's CREAM solar reaches condo residents.** CREAM roofs send their power to SDP's Innovation Park.

**CREAM cash leaks out.** Lease cash is spent anywhere; shared solar pays 9 sen/kWh and stops at 5 km.

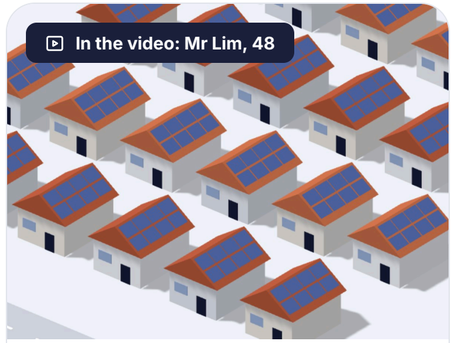
**Who is affected** condo residents & renters · roof hosts · EV drivers & would-be buyers · B40 & elderly · Sime Darby

**Why now** CREAM pilot in Elmina since Sep 2025 · 30,000-charger target by 2030 · SDP −40% Scope 1 & 2 by 2030

GENERATE SHARE USE

Source: SDP, Sustainability Report 2024, p.42 (May 2025) · SDP, Residential solar launch under NETR (Sep 2025) · DOSM, Living Quarters Special Report, Census 2020 (Apr 2026) · Energy Commission, Solar ATAP Guidelines (2026) · Kenanga Research, EV Day 2026 (Oct 2026) · Business Today, 6,904 public charging bays (Sep 2026) · Reuters/GSO via The Edge (Jan 2026) · Bernama TV, ~1 million households past 600 kWh (Sep 2026) · Yong et al., IJERPH 20(10):5910 (2023) · Energy Commission, SAC/CAC reduced by up to 40% (Aug 2025) · SDP, Net zero by 2050 pledge (Nov 2023)

# Five groups, one gap: solar that people can actually use



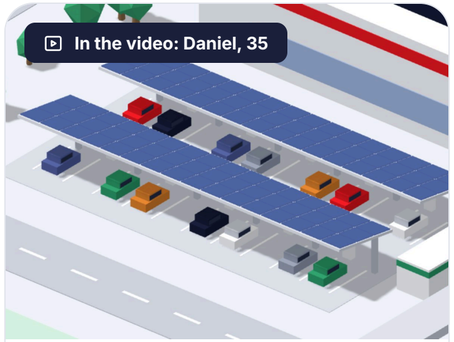

## SD landed owners
CREAM roof hosts

- Lease their roof, never use a kWh of it
- **Earn RM 0.10/kWh as credits (+20%) or a bill rebate**




## Condo residents & renters
no roof of their own

- No roof, no panels, no charger space
- **1 kW solar slice ≈ RM 40 off a month; balcony kit from 2027**



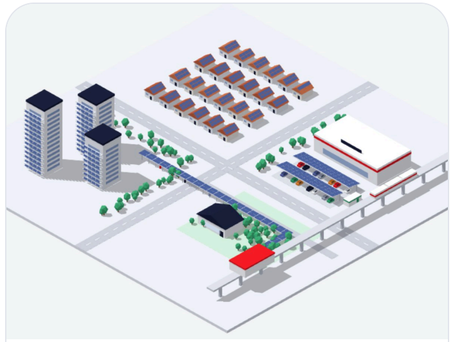

## Public EV drivers
mall visitors, commuters

- 18 battery EVs per bay; DC RM 0.88–1.80
- **Green-bay map, Saver RM 20 → RM 30, surplus discount**



## Future EV buyers
residents still on petrol

- Charging anxiety blocks the buy
- **Solar bays where they live + low running cost**



## Sime Darby Property
platform owner

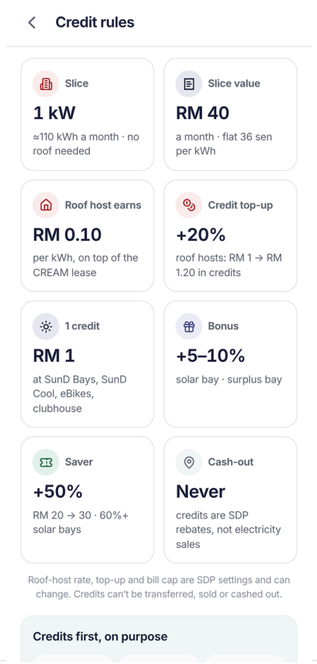
- Solar, chargers and storage not yet one loop
- **New revenue, measured ESG, a USP for every launch**

**Beyond drivers:** SunD eBike, Walk and Cool bring in kids, families and seniors, like **Uncle Tan, 72**.

GENERATE | SHARE | USE

Source: SDP, Residential solar launch under NETR (Sep 2025) · Kenanga Research, EV Day 2026 (Oct 2026) · EV Sifu, EV charging cost Malaysia 2026 · DOSM, Living Quarters Special Report, Census 2020 (Apr 2026) · SDP, City of Elmina

# Two core products, one loop: SunD Wallet shares the solar, SunD Bay spends it




CORE 1 · SUND WALLET APP

## Owners earn, renters save

### Owners earn on every kWh

RM 0.10/kWh on top of the lease, ≈ RM 50 a month. **Their choice:** credits +20% or a bill rebate (≤50%).

### Renters get a solar slice

A 1 kW slice ≈ **RM 40 off** a month, no roof needed. Funded by bay charging profit.

### The public tops up

Saver Credits: pay RM 20, get **RM 30** at solar bays and eBikes.




CORE 2 · SUND BAY

## Solar-first charging, open to all

75 kWp canopy | 180 kWh battery | AC + DC chargers | Grid tops up | eBike dock

**RM 1.40/kWh** (market 0.88–1.80) · **≈ RM 0.93** with Saver · cheaper when the battery is full.

AROUND THE CORE

### SunD Walk

Solar-roofed shade on the hottest route: −3.1 °C

### SunD Cool

Heat alert → nearest solar-cooled room, free; B40 & elderly first

### SunD eBike

Ride while you charge; credits pay

### SunD Balcony

800 W plug-in kits for condos, 2027

GENERATE | SHARE | USE

Source: SDP, Sustainability Report 2024, p.42 (May 2025) · PVGIS SARAH3 / NASA POWER via Sunveyor: KL 1,402 kWh/kWp/yr · RinggitPlus, Your new TNB bill explained (EEI table) · Trexon, Solar carport: ~2.5 kWp per bay (2026) · EV Sifu, EV charging cost Malaysia 2026 · Xu, Bardhan et al., Sci. Total Environ. 958 (2025) · The Star/Bernama, EC regulatory sandbox 2027 (Jun 2026)

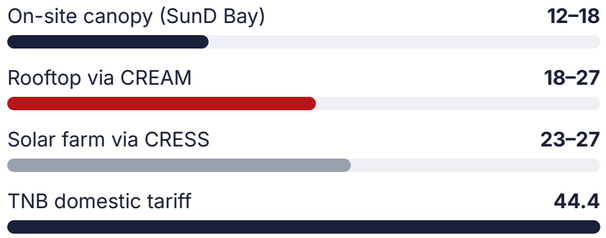
# Not another charging network: a closed solar loop residents live inside

| | TODAY | WITH SUND |
|---|---|---|
| **Reward** | Lease cash, spent anywhere | **Credits (+20%) or bill rebate, spent on green** |
| **Who's in** | Landed roof owners only | **Condo renters, B40, public drivers too** |
| **Energy path** | Shared solar: 5 km cap + 9 sen/kWh charge | **On-site canopy: 0 km, no access charge** |
| **Midday solar** | Not linked to residents | **Stored + surplus discount + 11am–3pm nudges** |
| **Power** | Public bays ~92% fossil | **Solar first, grid tops up** |
| **Heat** | No provision | **SunD Walk shade + SunD Cool rooms** |
| **Build** | Fixed sites | **Modular canopies, moved as demand grows** |

OUR EDGE

**Turns CREAM rooftop value into SunD Credits, while SunD Bays charge on their own sun first, anywhere.**

COST TO DELIVER 1 KWH · SEN

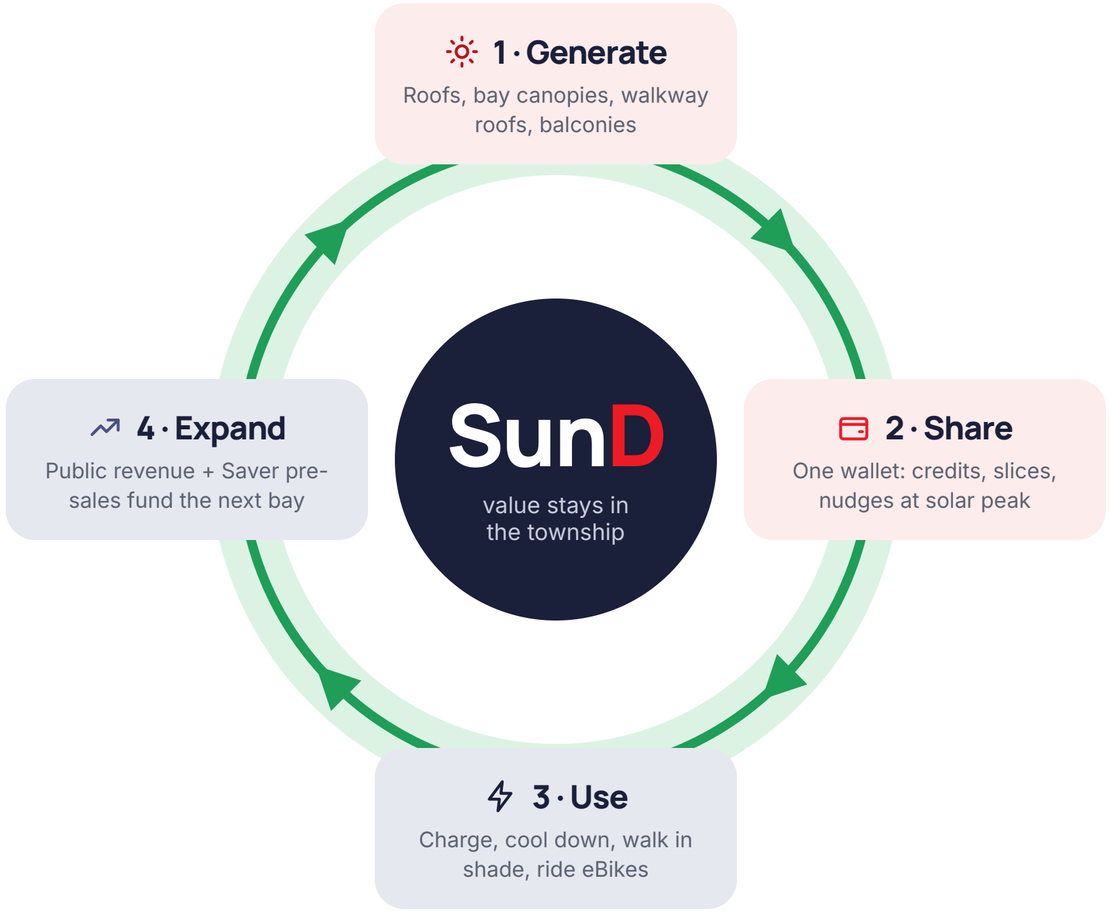



First in Malaysia: no scheme lets a roofless resident earn and spend solar on site.

GENERATE · SHARE · USE

Source: SEDA, Malaysia's 1st pilot run of P2P energy trading (2020) · Energy Commission, SAC/CAC reduced by up to 40% (Aug 2025) · The Edge, CRESS SAC now 18 sen/kWh (Oct 2026) · SoyaCincau, TNB domestic tariff structure July 2025 · Cobler, Commercial rooftop solar pricing guide (Aug 2026) · Reuters/GSO via The Edge (Jan 2026)

# One loop: generate → share → use → expand




GENERATE | SHARE | USE

TWO ENGINES, ONE WALLET

### CREAM engine
Rooftops keep selling to Innovation Park; owners earn credits.

### SunD Bay engine
Canopy + battery charge any driver, anywhere.

WHAT KEEPS IT SPINNING

### The battery sets the price
Full battery → Solar Surplus Discount; grid only tops up

### Nudges at solar peak
11am–3pm: pre-cool, run the washer, charge the car

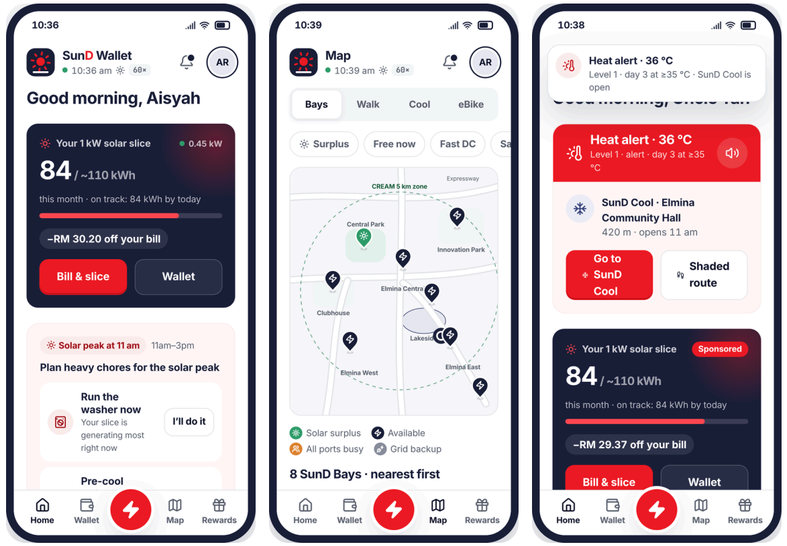
### The public pays less in the sun
Map points to the sunniest bay; Saver +50%

One bay lot: **~2.5 kWp ≈ 9 kWh a day ≈ 60 km** of e.MAS 5 driving, made and used on the same site.

Source: Trexon, Solar carport: ~2.5 kWp per bay (2026) · PVGIS SARAH3 / NASA POWER via Sunveyor: KL 1,402 kWh/kWp/yr · Carz, Proton e.MAS 5 launch (Oct 2025) · SDP, Residential solar launch under NETR (Sep 2025)

# Life in a SunD township: one Elmina precinct, drawn as the loop




Real screens from the working SunD Wallet prototype: home, green-bay map, heat alert.

TRY THE PROTOTYPE

**chchia28.github.io/SD-Hackathon**

+ a 4-minute 3D walkthrough video

1. CREAM homes · roof hosts earn
2. SunD Bay · mall car park
3. Condos · slices + balcony kits
4. SunD Walk · shaded route to bus stop
5. SunD Cool · community hall
6. Bus stop / transit hub

GENERATE | SHARE | USE

Source: SDP, Sustainability Report 2024, p.42: 228 homes, 1 MWp at Elmina Ridge 1 · SDP, City of Elmina: 5,000 acres, 36,600 homes

# Every side wins: residents save, streets cool, Sime Darby earns

### Residents

≈ RM 40 a month off with a 1 kW slice, no roof; shade and cool rooms on heat-alert days

### Roof owners

≈ RM 50 a month on top of the lease, as credits (+20%) or bill rebate

### Sime Darby

+RM 593 a day from a 30-lot mall bay, before credits; a USP every launch can print

### Planet

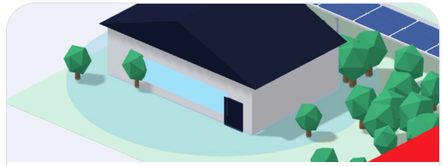
~73 t $CO_2$ a year avoided per medium bay; 0 km transmission

**RM 40/mo**
off the bill for a home with no roof

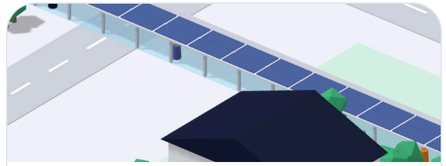
**−3.1 °C**
feels-like under SunD Walk shade

**−40%**
SDP Scope 1 & 2 by 2030: every bay reports into it

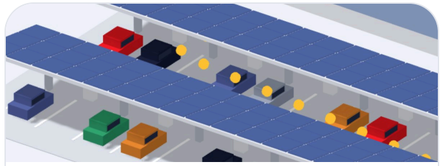
**36,600**
homes in City of Elmina alone, then every SD township



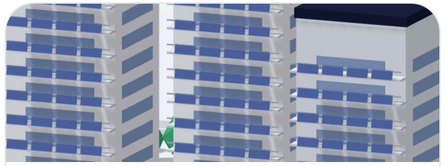
**Seniors**
SunD Cool rooms on heat-alert days



**Families & kids**
Shaded walks, eBike rides



**Public drivers**
Cheaper charging, Saver +50%



**Condo residents**
Balcony kits from 2027

**Reach grows**
Residents → public → other SD townships → other developers

GENERATE | SHARE | USE

Source: RinggitPlus, Your new TNB bill explained (EEI table) · PVGIS SARAH3 / NASA POWER via Sunveyor: KL 1,402 kWh/kWp/yr · Xu, Bardhan et al., Sci. Total Environ. 958 (2025) · Energy Commission, Grid Emission Factor 2022–2024 (Feb 2026) · SDP, Net zero by 2050 pledge (Nov 2023) · SDP, City of Elmina: 5,000 acres, 36,600 homes

# Commercial value: the cheapest kWh in town, sold at market price

## WHO PAYS, WHO EARNS

one SunD Wallet · credits stay in the township

| | | |
|---|---|---|
| Public drivers | pay | **RM 1.40 / kWh** list price, mid market · ≈ RM 0.93 with Saver Credits (A15) |
| Roof owners | earn | **≈ RM 50 a month** on top of the CREAM lease · ≈ RM 58 as credits · charge at RM 0.90 (A12, A20) |
| Renters | save | **≈ RM 40 a month** off the bill with a 1 kW slice, funded by bay profit (A11, A20) |

**SDP keeps the margin:** solar made on site costs 12–18 sen a kWh; public DC sells at RM 0.88–1.80 (A14, A15)

## ONE MALL BAY · 30 LOTS

capex **RM 1.02–1.47M** after GITA (A13)

| SESSIONS / LOT / DAY | NET CASH A YEAR | PAYBACK |
|---|---|---|
| 1.5 | RM 165k | 6.2–8.9 yrs |
| 2.0 | RM 216k | 4.7–6.8 yrs |
| 2.5 | RM 269k | 3.8–5.5 yrs |

Base case in red: **+RM 593 a day**. Assumes 18 kWh a session, 70% public use, RM 1.40 sale, RM 0.45 grid power (A14, A18); before Wallet credits and Saver discounts (A20). Panels last 25 years.

### Public charging

The cheapest kWh in the township, sold at a fair RM 1.40

### Prepaid Saver packs

RM 20 → RM 30 packs bring cash in before a bay is busy

### Credits spent in SD

Malls, clubhouse, eBikes, SunD Cool: spend and footfall stay local

### Sells homes

93% of Malaysians are willing or open to pay more for a green home (A2)

### Licensing

License SunD to other developers, malls and highways (A16)

## SCALE

**Not limited to SDP**

**Elmina pilot**
1 precinct · 10-lot bay

› **SD townships**
Bukit Raja, Serenia, more

› **Malls · offices · campuses**
any site, own canopy

› **Highways · petrol stations**
charge on the go

› **Other developers**
license SunD

**≈ RM 2.2M**
net a year from 10 mall bays, before credits

GENERATE | SHARE | USE

Source: EV Sifu, EV charging cost Malaysia 2026 · Cobler, Commercial rooftop solar pricing guide (Aug 2026) · Trexon, EV charger installation cost 2026 (Mar 2026) · MGTC/MyHIJAU, GITA asset list (Apr 2025) · Business Today, PropertyGuru 1H26 Consumer Sentiment Study (Sep 2026) · SDP, City of Elmina · Workings and assumptions: Appendix A11–A18

# Realistic today: Sime Darby already owns the hardest pieces

## Already in place

- **CREAM pilot in Elmina** (Sep 2025): SDP is the aggregator; Innovation Park buys the power
- **Solar car parks are allowed**: Solar ATAP at single-owner sites (s.7.2); zero-export self-use at malls (A21)
- **Credits are SDP rebates**, like a loyalty scheme: no new licence
- **SD-owned malls, parks, clubhouses** are the bay sites

**4.7–6.8 yrs**
payback, mall bay, before credits

**+RM 593**
net cash a day, before credits (A20)

**100% → 60%**
GITA on chargers (2027 proposed)

**3–4 mo**
to build a medium bay

**Honest trade-off:** the 10-lot pilot measures; the mall bay pays back (A14).

WEEKS 1–8
### Build
SunD Wallet live for one Elmina precinct; 10-lot pilot bay

MONTHS 3–6
### Pilot
7 KPIs measured; go / no-go at month 6

MONTHS 6–12
### Extend
30-lot mall bay, open to public; Saver Credits; first SunD Walk

YEAR 2+
### Scale
SunD sub-brand: other townships, highways, eBike; license to others

**Demand risk: managed** Built-in credit users · Saver pre-sales = cash upfront · Modular canopies move to busy sites · Stakeholders: SDP · TNB/GSPARX · residents · installer

GENERATE SHARE USE

Source: SDP, Residential solar launch under NETR (Sep 2025) · SDP, Sustainability Report 2024, p.42 (May 2025) · Energy Commission, Guidelines for Solar ATAP (in force 1 Jan 2026) · MGTC/MyHIJAU, GITA asset list (Apr 2025) · SoyaCincau, Budget 2027 cuts EV-charging GITA to 60% (Oct 2026) · Trexon, EV charger installation cost 2026 (Mar 2026)

# SunD connects what Sime Darby already owns into one loop, and proves it in one pilot

## Why SunD deserves to win

SunD turns Sime Darby's rooftops and car parks into one solar loop. Owners earn, renters save with no roof, and the public charges cheaper on solar. Credits also pay for eBikes, cool rooms and shaded walks, so demand is built in. It runs on schemes in force today, pilots in one Elmina precinct, and the prototype already works.

**Live, working prototype**
SunD Wallet app + SDP dashboard, in the browser

**4-minute 3D walkthrough**
Every screen in it is the real app

**Every number sourced**
Assumptions listed in appendix A18

**Pilot-ready in 6 months**
One precinct, seven KPIs, a clear go / no-go

## Your roof earns it. Your car drives it. Your township lives it.

**chchia28.github.io/SD-Hackathon**
Team UrbanByte · thank you
Appendix follows →

GENERATE | SHARE | USE

Source: SDP, Residential solar launch under NETR (Sep 2025) · Energy Commission, Guidelines for Solar ATAP (in force 1 Jan 2026)